MILITARY SPECIFICATION

CLOTH, PLAIN WEAVE, NYLON: WATER REPELLENT, OG-106

This interim amendment is approved for use within the Department of the Army, with MIL-C-43128C dated 10 September 1986.

PAGE 1

1.1, after "one type of" add "durable water repellent treated"

PAGE 2

Add the following:

"2.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the document which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICA ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies should be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)"

PAGE 3

3.4: Delete and substitute the following:

"3.4 Color. The color of the dyed and finished cloth shall be Camouflage Green 483 or Olive Green 106 and shall match the standard sample (see 6.5)."

3.4.1: Delete and substitute the following:

"3.4.1 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 $\pm$ 200 K, with illumination of 100 $\pm$ 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 $\pm$ 200 K."

AMSC N/A FSC 8305

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3.4.2: Delete and substitute the following:

"3 4.2 Colorfastness. The finished cloth shall show fastness to weathering and laundering equal to or better than the standard sample or equal to or better than a rating of "good". The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating not lower than 3.5. Testing shall be as specified in 4 4.3."

Add the following.

"3.4.3 Spectral reflectance. Cloth dyed Camouflage Green 483 shall conform to the reflectance specified below when tested as specified in 4.4.3.

Wavelength	Camouflage Green 483 Reflectance values (%)	
Nanometers (nm)	Min.	Max.
600	3	10
620	3	10
640	3	10
660	3	11
680	3	13
700	4	28
720	5	40
740	7	52
760	11	60
780	17	64
800	24	67
820	32	70
840	37	71
860	40	73"

PAGE 4

3.6.2· Delete and substitute the following:

"3.6.2 Water repellency. The cloth shall be given an approved fluorocarbon type water repellent treamtment (see 6.10) and shall conform to the requirements of 3.6.2.1, 3.6.2 2 and 3.6.2.3 "

3.6 2 1 Delete and substitute the following.

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PAGE 4 cont'd

"3.6.2.1 Spray rating. The results of the three individual determinations on the sample unit shall have an initial spray rating equal to or better than 100, 100, 90, and a spray rating after 3 launderings of 80, 80, 70, when tested as specified in 4.4.3."

Add the following:

"3.6.2.2 Hydrostatic pressure. The cloth shall show a hydrostatic height not lower than 40 cm for the individual sample unit and not lower than 45 cm for the lot average initially and after 3 launderings, when tested as specified in 4.4.3.

3 6.2 3 Resistance to organic liquids. The cloth shall show no wetting by n-tetradecane after 30 seconds initially and after 3 launderings, when tested as specified in 4.4.3."

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Table II, under "Colorfastness to:" : Add:

"Crocking	3.4.2	5651"

Under "Spray rating": Add:

"Initial	3.6.2.1	5526
After 3 launderings	3.6.2.1	5556 5/, 5526"

Add the following:

"Spectral reflectance	3.4.3	4.5.3
Hydrostatic height:		
Initial	3.6.2.2	5514
After 3 launderings	3.6.2.2	5556 5/, 5514
Resistance to organic liquids:		
Initial	3.6.2.3	4.5.2
After 3 launderings	3.6.2.3	5556 5/, 4.5 2"

Under table, in 2/: Delete "80 hours" and substitute "40 hours".

Under table: Add the following:

"5/ The cotton procedure shall be followed. Complete wash and dry cycles shall be performed."

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Add the following:

"4.5.2 Resistance to organic liquid test. Place a small specimen of the cloth on smooth horizontal surface, face up. Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After 30 seconds examine the specimen under light at an angle. Absence of light reflectance at the cloth drop interface shall be taken as evidence of wetting. Three specimens (or areas) taken at various locations across the sample unit shall be tested. Evidence of wetting on one or more specimens shall be considered a test failure.

4.5.3 Spectral reflectance test Spectral reflectance data shall be obtained from 600 to 860 nm (see 6.8) relative to a barium sulfate standard, the preferred white reference standard. Other reference white standards may be used provided they are calibrated to an absolute white; i.e., Halon, magnesium oxide, or vitrolite tiles (see 6.9). The spectral band width at 860 nm shall be less than 26 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65 in the visible spectrum. The specular component of reflection shall be included in the measurement. Specimens shall be measured as a single layer of fabric backed by six layers of the same fabric. Readings will be taken on a minimum of two different areas and the data averaged. The specimen shall be viewed at an angle no greater than 10 degrees from normal. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy shall be within 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25-inch diameter. Failure of the near-infrared spectral reflectance requirements will occur when spectral reflectance limits are exceeded at four or more specified wavelengths "

6.1, after "canteen": Add "and sleeping bag stuff sack."

PAGE 11

6 5 Delete and substitute the following:

"6.5 Dyestuff formulation. A suggested, but not mandatory dyestuff formulations as follows:

Olive Green 106	Camouflage Green 483
Acid Orange 156	Acid Orange 162
Acid Red 266	Acid Blue 171"
Acid Blue 258	
Acid Yellow 219	

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6.5, after "olive green". Add "and Camouflage Green 483".

6.5, above existing formulation: Add heading "Olive Green".

6.5: Add formulation for Camouflage Green 483 as follows:

Camouflage Green 483

Acid Orange 162
Acid Blue 171

Add the following:

"6.8 White standard. Barium sulfate of suitable quality for use as a white reference standard is available from the Eastman Kodak Company. The same source has available magnesium reagent (ribbon) and Halon. Suitable tiles can be obtained from the National Bureau of Standards or the instrument manufacturers.

6.9 Spectrophotometers. Suitable spectrophotometers for measuring spectral reflectance in the visible/near-infrared are the Diano Hardy, Diano Match Scan, Hunter D54P-IR, Hunter VIS/NIR Spectrocolorimeter, and Macbeth 1500 with IR option.

6.10 Approval of fluorcarbon water repellent Approval of the fluorocarbon type water repellent treatment is the responsibility of the U.S. Army Natick Research, Development and Engineering Center (GL), Natick, MA 01760-5014 and is based on extensive tests, including those for toxicity, which are not set forth in this document. Because of the time necessary to conduct full evaluation, only those chemical treatments already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement."

Custodians:

Army - GL
Navy - NU

Preparing activity:

Army - GL

Project No. 8305-A274

Review activities:

Army - MD
Navy - MC
DLA - CT

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